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Q.1

State the Cauchy's Theorem? ✓

Ans:-

Let "p" be a prime. Let G be a finite group and let "p" divide |G|. Then G has an element of order "p" and consequently, a subgroup of order p.

Q.2* (2021)

Write down the elements of $1+5\mathbb{Z}$? ✓

Sol:-

We Know that set of integers

$$\mathbb{Z} = \{0, \pm 1, \pm 2, \pm 3, \pm 4, \cdots\}$$

$$\mathbb{Z} = \{\cdots, -4, -3, -2, -1, 0, 1, 2, 3, 4 \cdots\}$$

$$5\mathbb{Z} = \{\cdots, -20, -15, -10, -5, 0, 5, 10, 15, 20 \cdots\}$$

$$1+5\mathbb{Z} = \{\cdots, (1+(-20)), (1+(-15)), (1+(-10)), (1+(-5)), (1+0), (1+5), (1+10), (1+15), (1+20) \cdots\}$$

$$1+5\mathbb{Z} = \{\cdots, -19, -14, -9, -4, 1, 6, 11, 16, 21, \cdots\}$$

Answer.

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Q-3

Define Factor or quotient group? ✓

Ans:-

Let $(H, *)$ be a normal subgroup of the group $(G, *)$. $(G/H, \otimes)$ is called factor or quotient group, where the operation $\otimes$ is defined on G/H by

$$Hg_1 \otimes Hg_2 = H(g_1 * g_2)$$

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Q-4

Write down all the generating sets of the Klein 4-group $V = \{e, a, b, c\}$?

Sol:-

The generating sets of the Klein 4-group $V = \{e, a, b, c\}$ are,

i) $\{a, b\}$ ii) $\{a, c\}$

iii) $\{b, c\}$ iv) $\{a, b, c\}$

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Q.5 ✓

Show that the homomorphism $h: G \to G'$ is injective if and only if $\ker h = \{e\}$.

Proof:-

Let h is injective and $x \in \ker h$.

then $h(x) = e' = h(e)$.

Hence $x = e$

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Conversely,

Let $\ker h = \{e\}$ and $x, y \in \ker h$

then,

$h(x) = h(y)$

$h(xy^{-1}) = h(x)h(y^{-1})$

$h(x)h(y)^{-1} = e'$

Since $xy^{-1} \in \ker h$

$xy^{-1} = e$

$x = y$

Hence,

h is injective.

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Q-6

Let G be a group. The set of all commutators $aba^{-1}b^{-1}$ for $a, b \in G$ generates a subgroup C of G?

Proof:-

Let $a, b \in G$

then,

$(aba^{-1}b^{-1})(aba^{-1}b^{-1})^{-1}$

$= aba^{-1}b^{-1}bab^{-1}a^{-1}$

We know that $aa^{-1} = e$

$= e \in C$

So, $e = eee^{-1}e^{-1}$

e is a commutator.

Q-7

Define Kernel of homomorphism?

Ans:-

Let $h: G \longrightarrow G'$ be a homomorphism of groups. The subgroup

$$h^{-1}[\{e'\}] = \{x \in G \mid h(x) = e'\}$$

is the Kernel of homomorphism. And denoted by $Ker(h)$.

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Q.8

Let H and K be normal subgroups of a group G with $K \leq H$. Then show that $G/H \simeq (G/K)/(H/K)$?

Proof:-

Let $\phi : G \to (G/K)/(H/K)$ be given by $\phi(a) = (aK)(H/K)$ for $a \in G$.

Clearly ϕ is onto $(G/K)(H/K)$, and for $a, b \in G$,

$$\phi(ab) = [(ab)K](H/K)$$

$$= [(aK)(bK)](H/K)$$

$$= [(aK)(H/K)][(bK)(H/K)] = \phi(a)\phi(b)$$

So ϕ is a homomorphism.

The Kernel consists of those $x \in G$ such that $\phi(x) = H/K$.

These x are just the elements of H.

Then first isomorphism theorem shows that

$$G/H \simeq (G/K)(H/K)$$

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Q-9

Show that $Z(G)$ is a normal and an abelian subgroup of G?

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Sol:-

For each $g \in G$ and $z \in Z(G)$ we have

$$gzg^{-1} = zgg^{-1} = ze = z$$

We see at once that $Z(G)$ is a normal subgroup of G. It implies that

$$gz = zg \text{ for } g \in G \text{ and } z \in Z(G).$$

If G is abelian, then

$$Z(G) = G$$

In this case, the center is not useful.

Q-10

State the first theorem of Isomorphism?

Ans:-

Let $\phi : G \to G'$ be a homomorphism with Kernel K, and let $\gamma_K : G \to G/K$ be the canonical homomorphism. There is a unique isomorphism $\mu : G/K \to \phi[G]$ such that $\phi(x) = \mu(\gamma_K(x))$ for each $x \in G$.

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Q.11

If N is a normal subgroup of G, then G/N is abelian if and only if $C \leq N$. where C is a commutator subgroup?

Proof:-

If N is a normal subgroup of G and G/N is abelian, then

$$(a^{-1}N)(b^{-1}N) = (b^{-1}N)(a^{-1}N)$$

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that is, $aba^{-1}b^{-1}N = N$

So, $aba^{-1}b^{-1} \in N$ and $C \leq N$.

Finally, if $C \leq N$, then

$$(aN)(bN) = abN$$
$$= ab(b^{-1}a^{-1}ba)N$$
$$= (abb^{-1}a^{-1})baN$$
$$= ebaN$$
$$= baN$$
$$= (bN)(aN)$$

Answer.

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Q-12

Find the left and right cosets of $H=A_3$ and $K=\{(1),(12)\}$ in S_3?

Sol:-

We calculated the right cosets of $H=A_3$.

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Right Cosets.

$H=\{(1),(123),(132)\}$

$H(12)=\{(12),(13),(23)\}$

Left Cosets

$H=\{(1),(123),(132)\}$

$(12)H=\{(12),(23),(13)\}$

In this case, left and right cosets of H are same.

However, the left and right cosets of K are not same.

Right Cosets.

$K=\{(1),(12)\}$, $K(13)=\{(13),(132)\}$, $K(23)=\{(23),(123)\}$

Left Cosets.

$K=\{(1),(12)\}$, $(23)K=\{(23),(132)\}$, $(13)K=\{(13),(123)\}$.

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Q.13

Define Normal Group with example?

Ans:-

A subgroup H of a group is a normal subgroup if $gH = Hg$ for all $g \in G$.

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Example:-

- $H_4 = \{(1), (1,2,3), (1,3,2)\}$ are subgroups of S_3.
- H_4 is a normal subgroup.

Q.14

Define Canonical homomorphism?

Ans:-

A mapping $f: G \to G/H$ is a homomorphism, and call it canonical homomorphism.

Q.15

Define Commutator subgroup?

Ans:-

The set of all commutators $aba^{-1}b^{-1}$ for $a, b \in G$ generates a subgroup C of G is called the commutator subgroup.

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Q.16 * (2020)

Write down the elements of the cyclic subgroup H of $Z_4 \times Z_6$, which is generated by $(0,1)$

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Sol:-

$$G = Z_4 \times Z_6$$

$$H = \langle (0,1) \rangle = \{(0,1)^1, (0,1)^2, (0,1)^3, \ldots, (0,0)\}$$

Now,

$(0,1)^1 = (0,1)$

$(0,1)^2 = (0,1) + (0,1) = (0,2)$

$(0,1)^3 = (0,1) + (0,1) + (0,1) = (0,3)$

$(0,1)^4 = (0,1) + (0,1) + (0,1) + (0,1) = (0,4)$

$(0,1)^5 = (0,1) + (0,1) + (0,1) + (0,1) + (0,1) = (0,5)$

$(0,1)^6 = (0,1) + (0,1) + (0,1) + (0,1) + (0,1) + (0,1) = (0,0)$

So,

$$H = (0,1) = \{\langle H = \{(0,0), (0,1), (0,2), (0,3), (0,4), (0,5)\rangle\}$$

Answer.

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Q.17 For $a, b \in G$,

Let $(G, \cdot)$ be a group with subgroup H. What do you mean by a congruent to "b" modulo H?

M.m

Ans:-

For $a, b \in G$, we say that a is congruent to b modulo H, and write $a \equiv b \bmod H$ if and only if $ab^{-1} \in H$. The relation $a \equiv b \bmod H$ is an equivalence relation on G. The equivalence class containing a can be written as,

$$Ha = \{ha \mid h \in H\}$$

it is called right coset of H in G.

Q.18 * (2021)

Show that a factor group of a cyclic group is cyclic? ✓

Ans:-

Let G be a cyclic with generator a, and let N be a normal subgroup of G. We claim the coset aN generates G/N. We must compute all powers of aN. But this amounts to computing, in G, all powers of the representative a and all these powers gives give all elements in G. Hence the powers of aN certainly give all cosets of N and G/N is cyclic.

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Q.19 *(2020)

Compute the factor group $(Z_4 \times Z_6)/\langle(0,1)\rangle$.

Soln:-

$G = Z_4 \times Z_6$, $|G| = 4 \times 6 = 24$

$H = \langle(0,1)\rangle$, $H = \{(0,0), (0,1), (0,2), (0,3), (0,4), (0,5)\}$

$|H| = 6$

order of factor group $= \frac{|G|}{|H|} = \frac{24}{6} = 4$

Since $Z_4 \times Z_6$ is abelian, so is $(Z_4 \times Z_6)/H$.

In additive notation, the cosets are

$H = (0,0)+H, (1,0)+H, (2,0)+H, (3,0)+H$

Since we can compute by choosing the representative $(0,0), (1,0), (2,0), (3,0)$.

It is clear that $(Z_4 \times Z_6)/H$ is isomorphic to Z_4) and the second factor of $Z_4 \times Z_6$, Z_6 collapses, leaving just the first factor Z_4.

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Q.20 *(2020)

Let $GL(2,R)$ denotes the set of all invertible matrices over R. Show that $K_4 = \{e, a, b, c\}$ is a group under matrix multiplication?

Sol:-

Given $K_4 = \{e, a, b, c\}$

To have $GL(2,R)$ is a Group.

i) Closure property:-

since the product of 2X2 non-singular matrices is a 2X2 non-singular matric.

ii) Associative property:-

if A, B, C are 2X2 non-singular matrices, the $A(BC) = (AB)C$, as the matrix multiplication is associative.

iii) Inverse property:-

If $A \in GL(2,R)$, then A^{-1} exists, since A is non-singular therefore every element of $GL(2,R)$ has inverse.

All group axioms are satisfied in $GL(2,R)$.

Hence,

$GL(2,R)$ is a group.

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Q.21*(2019)

Show that an intersection of normal subgroups of a group G is again a normal subgroup of G?

Ans:-

Let H_i be normal subgroups of G (for $i \in I$ where I is an index set). Let $H := \bigcap_{i \in I} H_i$. Then for $g \in G$ by left and right cancelation laws and the normality of H_i we have

$$gH = \bigcap_{i \in I} (gH_i) = \bigcap_{i \in I} (H_i g) = \left(\bigcap_{i \in I} H_i\right) g = Hg$$

Q.22*(2019)

Given any subset S of a group G, Show that it makes sense to speak of the smallest normal subgroup that contains S?

M.m

Ans:-

Let $\{H_i \mid i \in I\}$ be the collection of all normal subgroups of G that contain S. $H := \bigcap_{i \in I} H_i$ is a normal subgroup of G. Obviously H contains S. So H is smallest normal subgroup that contains S.